

Machine Learning for Imaging

Mammographic density prediction

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1 Choose a sensible data augmentation pipeline

To improve generalisation while preserving clinically meaningful density information, I designed two training-time augmentation pipelines of increasing complexity. Both are applied only during training, with photometric transforms applied before geometric ones, after intensity normalisation but before greyscale-to-RGB channel replication.

Pipeline 1 (used with Model 1) applies conservative augmentations. **Photometric:** `ColorJitter` (brightness ± 0.10 , contrast ± 0.15 , $p=0.5$) simulates exposure variation across scanners and acquisition protocols. **Geometric:** `RandomAffine` ($\pm 15^\circ$ rotation, $\pm 5\%$ translation, scale 0.9–1.1) reflects patient positioning variability. Ranges were deliberately modest to avoid distorting clinically relevant density patterns.

Pipeline 2 (used with Model 2) extends this for greater robustness. Stronger `ColorJitter` (± 0.3 brightness and contrast) is combined with `GaussianBlur` (kernel 3, $p=0.3$) to mimic defocus or detector noise, and `RandomAdjustSharpness` (factor 2, $p=0.3$) for edge clarity variation. Geometrically, `RandomHorizontalFlip` ($p=0.5$) exploits left/right breast symmetry, `RandomAffine` ($\pm 10^\circ$, $\pm 5\%$ translation, scale 0.9–1.1) adds pose variation, and `RandomErasing` ($p=0.2$, scale 0.02–0.08) discourages reliance on localised tissue features.

I deliberately avoided vertical flips, as mammograms have a consistent top-to-bottom anatomical orientation. Together, these pipelines compare a minimal, clinically cautious strategy with a more comprehensive augmentation design.

2 Implement a sensible model for image classification

I evaluated two architecturally distinct models pre-trained on ImageNet-1K, fine-tuned for four-class density prediction. Pre-trained backbones provide low- and mid-level features that transfer well to greyscale mammographic images (replicated to three channels for compatibility).

Model 1: ResNet-18. Residual connections stabilise gradient flow during fine-tuning. I replaced only the final fully connected layer ($512 \rightarrow 4$) while retaining all pre-trained convolutional weights. Training: AdamW ($\text{lr}=10^{-3}$, weight decay 10^{-4}), batch size 256, 3 epochs.

Model 2: EfficientNet-B0. EfficientNet uses compound scaling to jointly optimise depth, width, and resolution, achieving better accuracy per parameter. It employs MBConv blocks with squeeze-and-excitation attention, producing a 1280-dimensional feature vector. I replaced the classifier head ($1280 \rightarrow 4$). Training: AdamW ($\text{lr}=10^{-4}$, weight decay 10^{-4}), batch size 64, 7 epochs. The lower learning rate avoids destabilising pre-trained weights.

Both models use `ImbalancedDatasetSampler` to address class imbalance ($\sim 10\%$ A, $\sim 43\%$ B, $\sim 42\%$ C, $\sim 5\%$ D), cross-entropy loss on raw logits, and `ModelCheckpoint` monitoring validation AUC.

Observations. Both models show steady training loss decrease and AUC increase, but validation performance peaks early (Figure 1). After the peak, validation loss rises while training loss continues to fall, suggesting onset of overfitting. ResNet-18 achieves best validation AUC (0.933) at epoch 3; EfficientNet-B0 peaks at epoch 7 (validation AUC 0.927). On the test set, ResNet-18 achieves macro

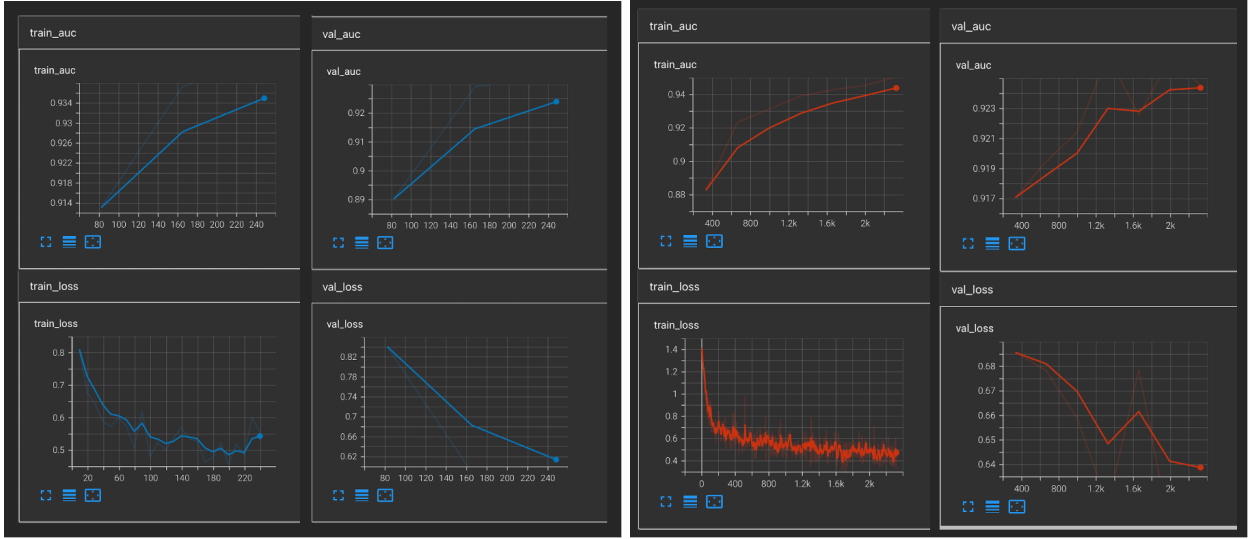


Figure 1: Training and validation curves. Left(in blue): ResNet-18 (3 epochs). Right(in orange): EfficientNet-B0 (7 epochs).

AUC 0.922 and EfficientNet-B0 achieves 0.917. Both models achieve strong test AUC above 0.91, confirming that ImageNet features transfer effectively to mammographic images even at low resolution (128×96).

3 Conduct a subgroup performance analysis

I analysed performance across five metadata attributes (Table 1). **Laterality** and **view** show near-identical AUC (≈ 0.92), confirming the model does not rely on laterality- or projection-specific features. **Age** performance is stable across bins (0.913–0.929) with no systematic bias. The key finding is **implant status**: the implant subgroup ($n=287$) shows a marked performance drop (macro AUC 0.810 vs 0.925), indicating a genuine gap likely caused by implants obscuring tissue patterns (Figure 2). **Cancer-positive** cases show higher AUC (0.952), but with only 129 samples and jagged ROC curves, this likely reflects variance-driven inflation rather than genuinely better discrimination.

Attribute	Subgroup	N	Macro AUC
Laterality	L/R	2936/2929	0.921/0.923
View	CC/MLO	2928/2937	0.924/0.921
Age	<50/50–60/60–70/70+	1963/1591/1506/800	0.929/0.921/0.913/0.924
Implant	No/Yes	5578/287	0.925/0.810
Cancer	No/Yes	5736/129	0.921/0.952

Table 1: Macro AUC by subgroup (ResNet-18, test set).

4 Model inspection

I extracted 512-dimensional embeddings from the ResNet-18 penultimate layer (before the classification head), reduced them to 4 components via PCA (capturing 97.3% of variance), then projected to 2D with t-SNE. PCA is applied first to reduce noise and improve t-SNE stability on high-dimensional input.

In the PCA projection (Figure 3), density classes arrange along a smooth ordered manifold ($D \rightarrow C \rightarrow B \rightarrow A$), suggesting the model encodes density as a continuous spectrum rather than discrete clusters. Substantial overlap between classes B and C is visible, consistent with their lower per-class AUC and the known subjectivity of this boundary among radiologists.

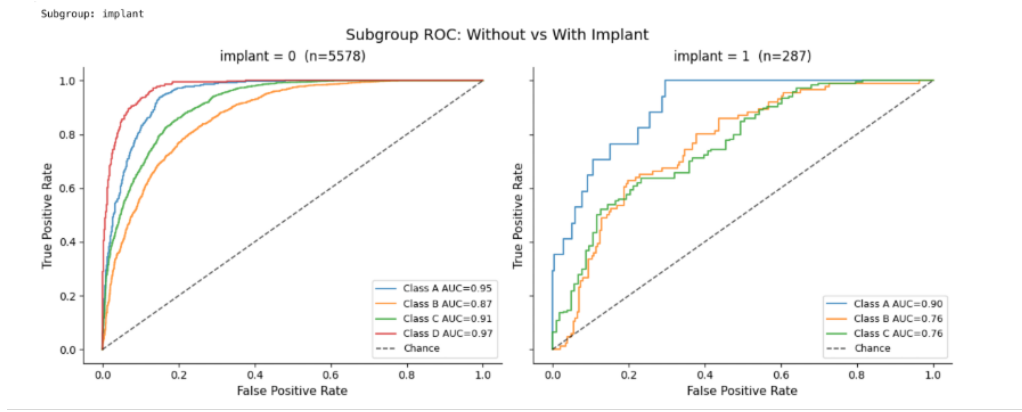


Figure 2: ROC curves by implant status, showing reduced performance for implant cases (n=287).

The t-SNE projection (Figure 4) reveals clearer local structure: extreme classes (A and D) form distinct regions while B and C remain intermixed. Misclassifications concentrate at class boundaries, indicating errors arise from intrinsically ambiguous samples. When coloured by laterality and view, points are well mixed, confirming invariance to acquisition factors. Implant cases appear partially localised, consistent with their degraded subgroup performance and suggesting a mild domain shift. Cancer-positive samples are sparse and do not form a distinct cluster.

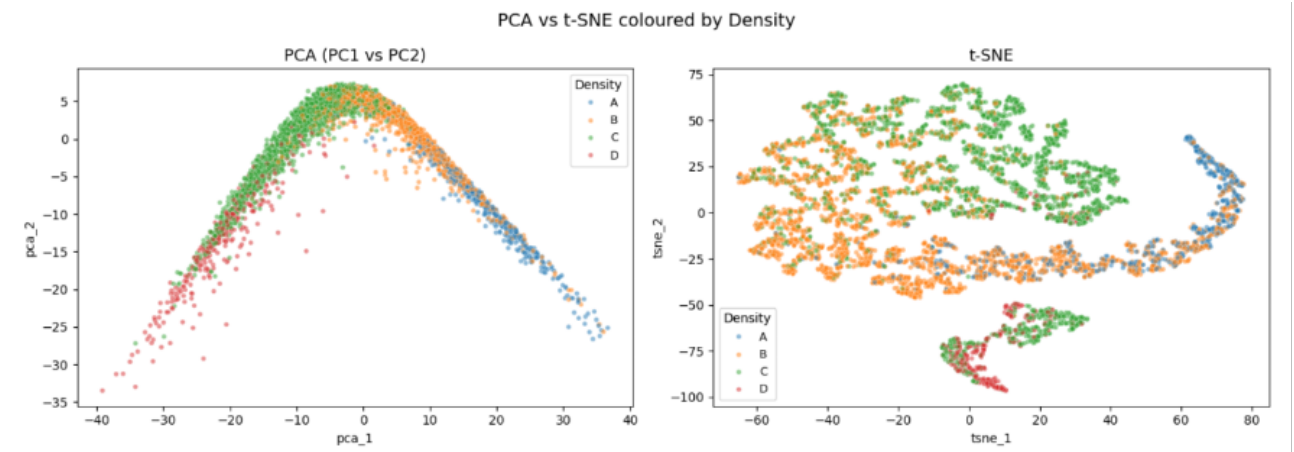


Figure 3: PCA and t-SNE projections coloured by true density class.

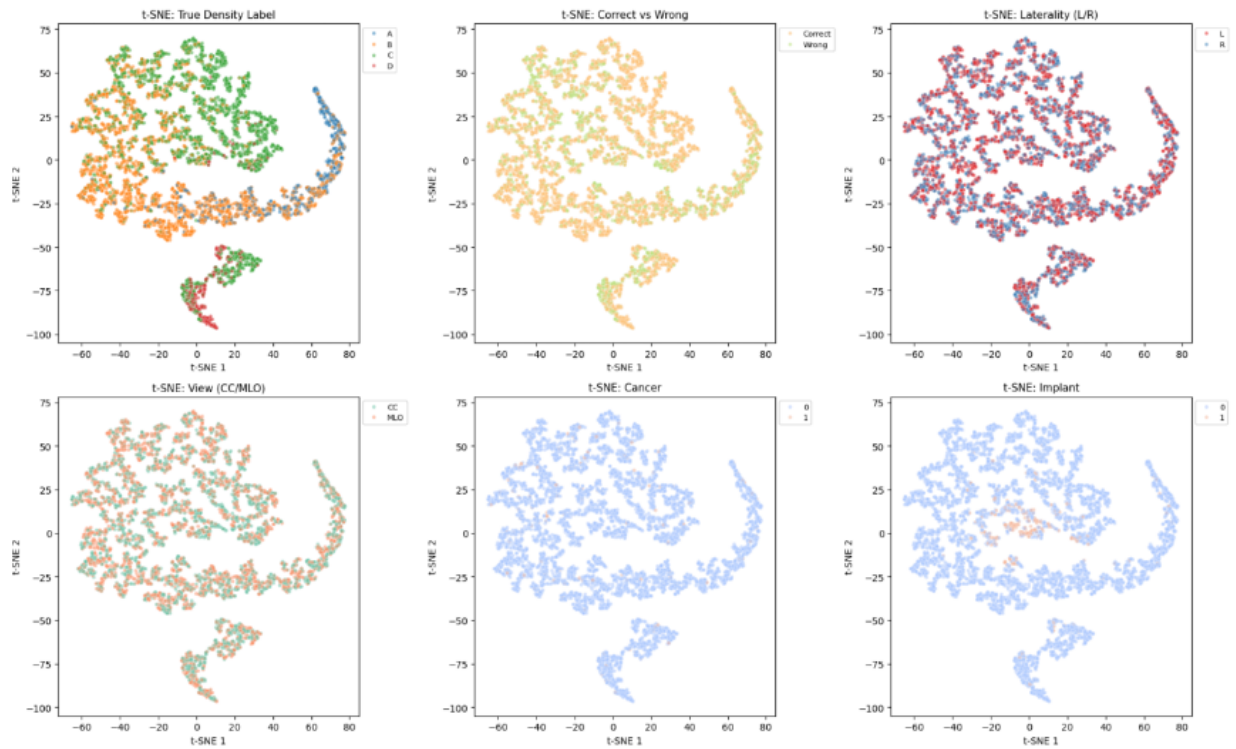


Figure 4: t-SNE projections coloured by density, prediction correctness, and metadata attributes.